

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1018

State Street 230 kV
October 24, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1018. The Interconnection Customer proposes a 1200.1 MW natural gas combined cycle facility interconnecting at the State Street 230 kV substation in the Indiana area. The request was received on June 27, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 378 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$78,241,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SEGS 5-STATE ST 100 kV
Contingency: RICHLAND-DEKALB 500 kV
Loading: 180.4% (Rating: 186 MVA)
DFAX: 8.1%
Pre-project loading: 127.8%
2. Facility: PEAVINE-STATE ST 100 kV Cir. 1
Contingency: DEKALB-RICHLAND 500 kV L
Loading: 154.9% (Rating: 155 MVA)
DFAX: 6.3%
Pre-project loading: 105.9%
3. Facility: PEAVINE-STATE ST 100 kV Circuit 2
Contingency: RICHLAND-DEKALB 500 kV
Loading: 154.9% (Rating: 155 MVA)
DFAX: 6.3%
Pre-project loading: 105.9%
4. Facility: STATE STREET SUBSTATION 230/100 kV TX
Contingency: RICHLAND-DEKALB 500 kV
Loading: 122.1% (Rating: 256 MVA)
DFAX: 7.6%
Pre-project loading: 86.6%
5. Facility: STATE ST SUBSTATION 230/100 kV Auto Bank 2
Contingency: RICHLAND-DEKALB 500 kV
Loading: 122.1% (Rating: 256 MVA)
DFAX: 7.6%
Pre-project loading: 86.6%

6. Facility: STATE STREET 230/100kV TX Bank 3
Contingency: RICHLAND SUBSTATION-DEKALB 500 kV
Loading: 122.1% (Rating: 256 MVA)
DFAX: 7.6%
Pre-project loading: 86.6%
7. Facility: STATE ST 230-100 kV Autotransformer
Contingency: DEKALB-RICHLAND 500 kV
Loading: 122.1% (Rating: 256 MVA)
DFAX: 7.6%
Pre-project loading: 86.6%
8. Facility: MONTE DEL ESTADO 6303-HARPER 100 kV
Contingency: RICHLAND-DEKALB 500 kV line
Loading: 119.3% (Rating: 308 MVA)
DFAX: 5.1%
Pre-project loading: 99.3%
9. Facility: HILLCREST-MUDOON MILL 100 kV
Contingency: DARLINGTON RD-WAYNE TRACE 100 kV line
Loading: 118.4% (Rating: 319 MVA)
DFAX: 7.1%
Pre-project loading: 91.8%
10. Facility: WABASH-MONTPELIER ELECTRIC GENERATING STA 500 kV line
Contingency: SABANERA-LONETREE 500 kV Line
Loading: 116.5% (Rating: 2,847 MVA)
DFAX: 8.9%
Pre-project loading: 112.8%
11. Facility: DEAN FOREST-EASTOWN RD 230 kV
Contingency: MUSKOGEE-SWAN VLY 500 kV
Loading: 116.0% (Rating: 846 MVA)
DFAX: 7.7%
Pre-project loading: 105.0%
12. Facility: MUSKOGEE-DEAN FOREST 230 kV
Contingency: SWAN VLY-MUSKOGEE 500 kV
Loading: 115.6% (Rating: 861 MVA)
DFAX: 7.7%
Pre-project loading: 104.8%
13. Facility: WABASH 500/230 kV Autotransformer
Contingency: DEKALB-RICHLAND 500 kV
Loading: 114.8% (Rating: 1,022 MVA)
DFAX: 21.9%
Pre-project loading: 89.0%
14. Facility: MONTPELIER ELECTRIC GENERATING STATION 500/230 kV Auto
Contingency: WABASH-MONTPELIER ELECTRIC GENERATING STATION 500 kV
Loading: 113.3% (Rating: 1,479 MVA)
DFAX: 13.9%
Pre-project loading: 102.0%
15. Facility: WOODBURN-MONROEVILLE SUBSTATION 100 kV
Contingency: STATE STREET-ROBISON PARK 230 kV line
Loading: 113.1% (Rating: 177 MVA)
DFAX: 5.2%
Pre-project loading: 77.7%
16. Facility: SAGE-WOODBURN SUBSTATION 100 kV
Contingency: STATE ST SUBSTATION-ROBISON PARK 230 kV Line

- Loading: 111.2% (Rating: 191 MVA)
DFAX: 5.2%
Pre-project loading: 78.4%
17. Facility: ALLEN-AEP PETTIT AVENUE 100 kV L
Contingency: SEGS 5-HILLCREST 100 kV line
Loading: 110.7% (Rating: 354 MVA)
DFAX: 6.1%
Pre-project loading: 90.0%
18. Facility: HEBO-BARRETT 230 kV
Contingency: WINTHROP-HEBO 230 kV
Loading: 110.4% (Rating: 873 MVA)
DFAX: 6.5%
Pre-project loading: 101.4%
19. Facility: HILLCREST-WAYNE TRACE 100 kV line
Contingency: MULDOON MILL-HILLCREST 100 kV
Loading: 110.1% (Rating: 265 MVA)
DFAX: 6.4%
Pre-project loading: 80.9%
20. Facility: MORENO TAP-EASTOWN ROAD 230 kV Line
Contingency: TREGO SUBSTATION-SWAN VLY 500 kV
Loading: 109.3% (Rating: 1,070 MVA)
DFAX: 6.2%
Pre-project loading: 102.3%
21. Facility: MONTE DEL ESTADO 6303-SAGE 100 kV
Contingency: STATE ST-ROBISON PARK 230 kV
Loading: 109.0% (Rating: 210 MVA)
DFAX: 5.2%
Pre-project loading: 79.2%
22. Facility: HILLCREST SUBSTATION-SEGS 5 100 kV line
Contingency: RICHLAND-DKALB 500 kV
Loading: 108.8% (Rating: 554 MVA)
DFAX: 9.4%
Pre-project loading: 88.5%
23. Facility: E METHUEN-SWAN VLY 500 kV line
Contingency: MONTPELIER ELECTRIC GENERATING STATION SUBSTATION-MUSKOGEE 500 kV
Loading: 108.3% (Rating: 2,718 MVA)
DFAX: 5.1%
Pre-project loading: 106.0%
24. Facility: WABASH-MONFORT 230 kV Line
Contingency: DEKALB-RICHLAND 500 kV line
Loading: 106.7% (Rating: 1,056 MVA)
DFAX: 20.4%
Pre-project loading: 83.6%
25. Facility: BURRIS PIT-JAMES DE YOUNG 230 kV
Contingency: TREGO-BEAVER 500 kV line
Loading: 106.6% (Rating: 1,392 MVA)
DFAX: 8.7%
Pre-project loading: 99.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SEGS 5-STATE ST 100 kV	Rebuild line with higher-capacity conductor	\$4,430,000	68% (\$3,034,000)
PEAVINE-STATE ST 100 kV Cir. 1	Rebuild line with higher-capacity conductor	\$2,188,000	57% (\$1,257,000)
PEAVINE-STATE ST 100 kV Circuit 2	Rebuild line with higher-capacity conductor	\$2,093,000	57% (\$1,202,000)
STAE ST 230 kV/100 kV XFMR T1	Replace transformer with higher-rated unit	\$8,671,000	55% (\$4,781,000)
STATE ST SUBSTATION 230/100 kV Auto Bank 2	Replace transformer with higher-rated unit	\$11,635,000	55% (\$6,416,000)
STATE STREET 230/100kV TX Bank 3	Replace transformer with higher-rated unit	\$7,087,000	55% (\$3,908,000)
STATE ST 230-100 kV Autotransformer	Replace transformer with higher-rated unit	\$12,593,000	55% (\$6,944,000)
MONTE DEL ESTADO 6303-HARPER 100 kV	Rebuild line with higher-capacity conductor	\$8,002,000	50% (\$4,010,000)
HILLCREST-MUDOON MILL 100 kV	Rebuild line with higher-capacity conductor	\$6,125,000	45% (\$2,739,000)
WABASH-MONTPELIER ELECTRIC GENERATING STA 500 kV line	Rebuild line with higher-capacity conductor	\$121,381,000	5% (\$6,586,000)
DEAN FOREST-EASTOWN RD 230 kV	Rebuild line with higher-capacity conductor	\$18,577,000	17% (\$3,215,000)
MUSKOGEE-DEAN FOREST 230 kV	Rebuild line with higher-capacity conductor	\$33,880,000	17% (\$5,863,000)
WABASH SUBSTATION 500/230 kV Auto	Replace transformer with higher-rated unit	\$31,175,000	14% (\$4,358,000)
MONTPELIER ELECTRIC GENERATING STA SUBSTATION 500-230 kV TX	Replace transformer with higher-rated unit	\$29,131,000	10% (\$3,069,000)
WOODBURN-MONROEVILLE SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$6,623,000	64% (\$4,242,000)
SAGE-WOODBURN SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$2,613,000	64% (\$1,674,000)
ALLEN SUBSTATION-AEP PETTIT AVENUE 100 kV line	Rebuild line with higher-capacity conductor	\$8,052,000	46% (\$3,682,000)
HEBO-BARRETT 230 kV	Rebuild line with higher-capacity conductor	\$13,788,000	8% (\$1,089,000)
WAYNE TRACE-HILLCREST 100 kV line	Rebuild line with higher-capacity conductor	\$1,091,000	57% (\$618,000)
MORENO TAP-EASTOWN ROAD 230 kV Line	Reconductor line	\$18,321,000	10% (\$1,921,000)
SAGE-MONTE DEL ESTADO 6303 100 kV	Reconductor line	\$1,759,000	64% (\$1,127,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HILLCREST SUBSTATION-SEGS 5 100 kV line	Reconductor line	\$1,408,000	22% (\$304,000)
E METHUEN-SWAN VLY 500 kV line	Reconductor line	\$115,910,000	3% (\$3,197,000)
WABASH-MONFORT 230 kV	Reconductor line	\$8,463,000	14% (\$1,182,000)
BURRIS PT-JAMES DE YOUNG 230 kV	Reconductor line	\$23,086,000	8% (\$1,823,000)
	Total allocated to project		\$78,241,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.